

Abridged Survey Protocol:
Nereocystis kelp canopy productivity data from BC Central Coast v1.3.0
Compiled by Ondine Pontier, August 2021
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This is an abridged version of the Nearshore Kelp Monitoring protocols available [here](#).

Sites

Monitoring of *Nereocystis luetkeana* canopy dynamics is conducted at 6 permanent sites and multiple (~28) additional non-permanent sites (Fig. 1, refer to site_metadata.csv) in the vicinity of Calvert Island, British Columbia, Canada.

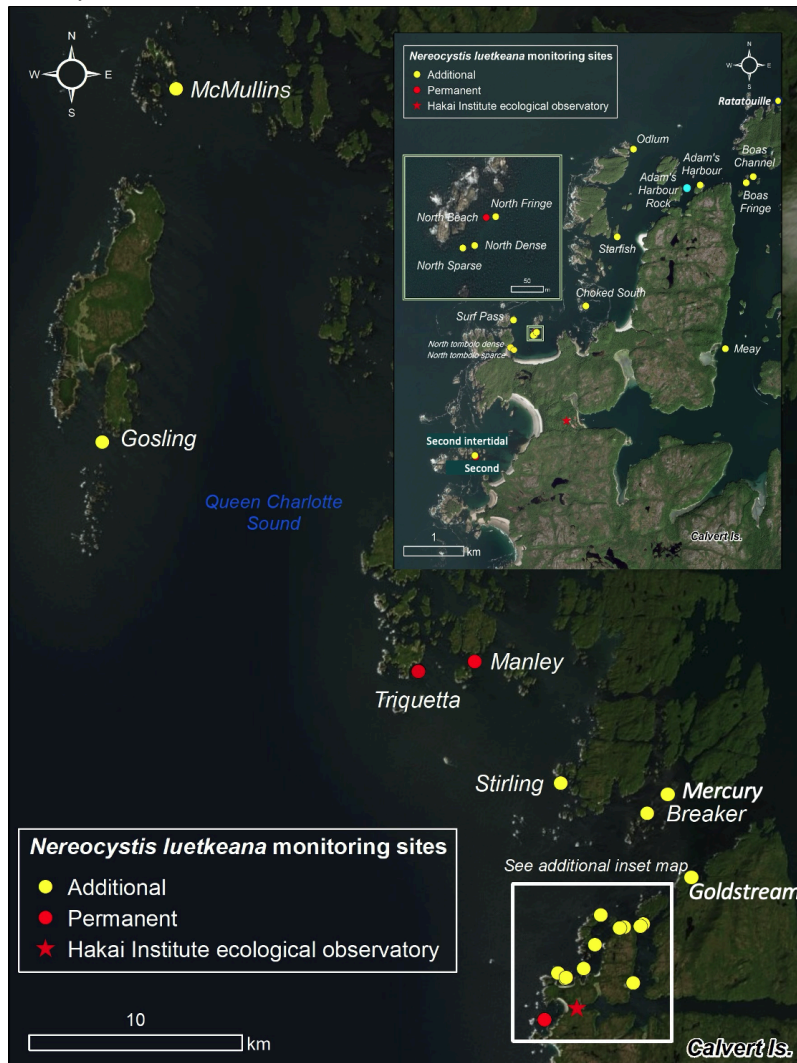


Figure 1. Map of site locations. Sites monitored in partnership with HIRMD (Heiltsuk Integrated Resource Management Department) do not appear on the map for privacy, though they can be made available upon request.

Protocols

The following field protocols are designed to capture measurements necessary for calculating *N. luetkeana* density (plants m^{-2}), biomass (kg m^{-2}), and productivity ($\text{kg m}^{-2} \text{d}^{-1}$). **Bold parameters** indicate notation found in the data and field spreadsheets, and all parameters are defined in the **data_dictionary**.

Site layout

Sites were selected on semi-exposed rocky reefs in beds dominated by *N. luetkeana* occurring within a depth range of 3-5 m below Sites were sampled at a range of intervals year round 2016-2019 with higher frequency during the growing season. 2019 onward, sites are visited on an annual basis. Permanent sites have three 20m leadlines placed parallel to one another and approximately 4m apart in the center of the kelp bed (Figure 1). Additional, non-permanent sites may have anywhere between 1 to 3 transects placed either parallel or in series and the length of the transects vary between 20 and 60m based on the site.

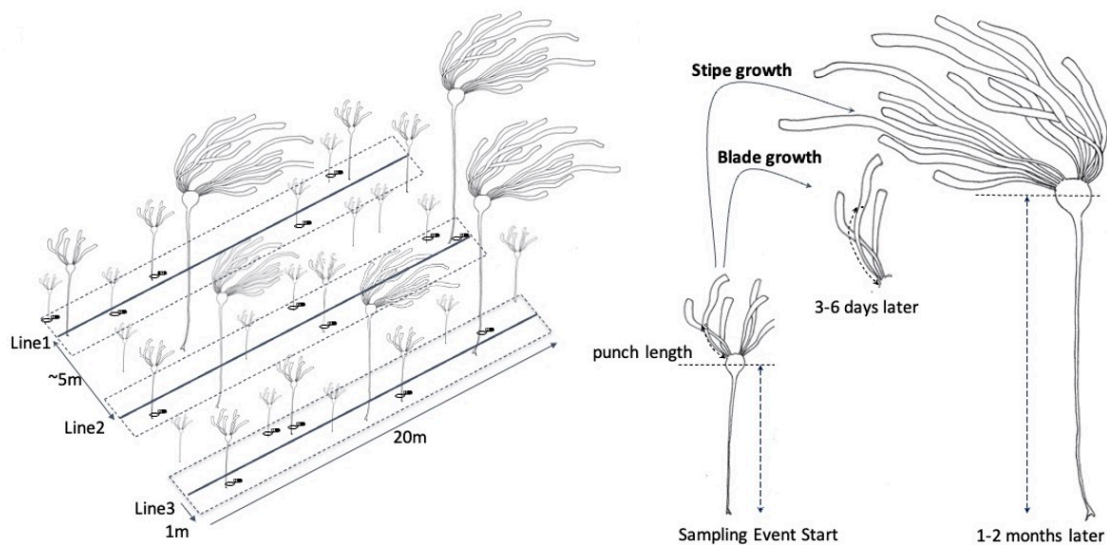


Figure 1. Site survey layout (left) including three permanent transects (bold lines), along which five individual kelps per transect were haphazardly tagged (circles and tags) and surveyed. Methods (right) of measuring and punching blades and measuring stipes after specific time periods to calculate growth.

Plant Density

Stipe density is the minimal amount of information that may be collected during a survey at a site. It is recorded in the **nereo_density.csv**. This includes any *N. luetkeana* adult (stipe >1m total length) within 1m of the transect on both sides of the transect (Fig 2. 2018 exceptions). Often 2 m increments (**bin**) are used to break up the transects.

Plant Size Structure

Plant size is generally collected in-situ by measuring the maximum sub-bulb diameter (**Sbulb_max**) of ~45 individuals randomly selected across the plot (along the 3 transects). This is recorded in **nereo_size.csv** (or in **nereo_metrics.csv** if additional parameters are collected). Prior to 2019, in-situ blade length, stipe length and blade count were collected as proxy for plant size rather than sub-bulb diameter. This is recorded in **nereo_metrics.csv**. In a subset of sites in 2018 and 2019 harvested plants were instead used as a representative sample of the bed and thus plant size information is directly inferred from the **nereo_harvest.csv**.

Growth Protocols (including growth measurements)

1. First timestamp (Metrics - T1)

A subsample of ~15 individuals are tagged, approximately 5 plants along each transect line. The following is done for each of the plants:

- a. Divers randomly select a plant as far as 1m on either side of the transect line, and mark the individual with a numbered tag. Divers record the tag number (**tag1**), and the distance of the tagged plant along the marked leadline. transect is recorded in the (**dist**) column.
- b. Measure stipe length (**stipeL1**), from the base of the holdfast to halfway up the pneumatocyst.
- c. Measure average blade length (**bladeL1**), from the pneumatocyst to the end of the bulk of the blades.
- d. Estimate percentage of blades that have reproductive sori. This is recorded as: 0%, ("0"), 0-50% ("L" - low), or 50-100% ("H" - high) in the **Psori1** column.
- e. Select an "outer" blade and measure its length (**obladeL1**) using a flexible tape measure. Outer blades are the blades attaching to the pneumatocyst on the outside edge of the blade attachment site.
- f. Punch a 3-5mm diameter hole at a distance that is approximately $\frac{1}{3}$ of the measured blade length. Punch distances are made relative to blade length to ensure that the majority of total blade elongation is captured during the measured growth interval (Kain et al. 1987; 95% of total blade growth occurs at the proximal third to half of the blade). Record punch distance (**distP1**) in cm. This is the distance from where the blade attaches at the pneumatocyst to the hole punch.

2. Second timestamp, 4-5 days later (Metrics - T2)

Relocate each tagged plant, find the punched blade and:

- a. Measure the distance from the pneumatocyst to the hole (**distP2**).

- b. Measure the punched blade total length (**obladeL2**), from the pneumatocyst to the end of the blade.
- c. Count and record the number of blades that are attached to pneumatocyst (**Bcount2**).
- d. Using calipers, measure the maximum sub-bulb diameter (**Sbulb_max2**). It is the width of the stipe at its widest point beneath the bulb and tightly correlated with total plant weight (Stekoll et al. 2004).

3. Third timestamp, approximately 1 month later (Metrics - T3)

For annual monitoring this timestamp can be anywhere between 2 weeks to a month after original tagging. In years when sites were monitored seasonally, this timestamp coincides with the T1 or T2 of the subsequent interval. Relocate the each tagged plant and measure the following, after which all tags are removed:

- a. Stipe length (**stipeL3**), from the base of the holdfast to halfway up the pneumatocyst.
- b. Blade count (**Bcount3**), the number of blades attached to pneumatocyst.

4. Density and Size Structure

The following is done for each of the 3 transect lines anytime between T1 and T2:

- a. Measure maximum sub-bulb diameter (**Sbulb_max**), stipe length (**stipeL**), average blade length (**bladeL**) and estimate sori percentage (**Psori**) on an additional 15 randomly selected individuals within the 2 m swath (1 m on each side of the transect line). Do not re-measure tagged plants (from T1) as these data are combined to increase sample size of above parameters. T
- b. Count all *N. luetkeana* adult (>1m total length) stipes within 1m of the transect on both sides of the transect. Record count every 2 m increments on both sides of the transect.

5. Harvest

Harvesting plants for biomass and allometric relationships involves collecting 20 plants at random from each site, once a year (July or August), outside of the monitoring plot and measuring the following on the dock or boat:

- a. Stipe length (**stipeL**) - measure from top of holdfast to center of pneumatocyst.
- b. Average blade length (**bladeL**) - measure from pneumatocyst to estimated length end, where most blades terminate.
- c. Pneumatocyst diameter (**pneuD**), widest part of the pneumatocyst .
- d. Sub-bulb diameter (**Sbulb_max**), width of the stipe at the widest point below the pneumatocyst.
- e. Blade count (**Bcount**).
- f. Blade width (**Bwidth**), width of randomly selected blade half way
- g. Blades wet weight (**bladeW**): cut the blades off at the top of the pneumatocyst and weigh them in a bucket using a fish scale.
- h. Stipe (whole) wet weight (**stipeW**) - weigh the remaining stipe in a bucket using the fish scale.
- i. Put aside a piece of stipe (~5cm) and blades (~5-50cm) for wet to dry regression.

6. Wet Dry

A piece of stipe and blade is kept and dried during harvest to obtain a wet to dry biomass conversion factor in the following manner.

- a. Empty trays are numbered with **sample IDs** and pre-weighed on an analytical balance (**tray**). A **sample ID** is allocated chronologically based on pre-existing data.
- b. An additional **unique ID** is allocated to the sample if it will be kept for isotope analysis. These are Nearshore-wide IDs and are kept track of on Nearshore Isotope files.
- c. The **tag** number for that plant is also recorded if known.
- d. The sample of either stipe or blade is measured to the nearest 0.1 cm (**length**) and recorded as well as the type of material (**stipe_blade**).
- e. The trays containing wet materials are weighed (**tray_wet**).
- f. The samples are dried in the drying oven overnight (12 hour minimum). Additional time may be required if the sample is not dry.
- g. The trays containing the dried material are weighed (**tray_dry**).

7. Plant turn-over

Additional plants were tagged to increase sample size in 2016 and 2017 after which it was discontinued. These data were then pooled with tag data extracted from metrics survey to estimate plant loss.

- a. At any time between timestamp 1 and 2:
 - i. Select 1-10 plants randomly and mark each with a zip tie. This may occur at any time between timestamp 1 and 2.
 - ii. Record the number of plants tagged (**new**). Do not tag already tagged plants.
 - iii. Record date and transect number and nearest distance along the leadline.
- b. Timestamp 3 (between timestamp 1 and 2 of subsequent interval):
 - i. Relocate plants and count how many remain (**found**).
 - ii. Take off zip ties.

8. Blade turn-over

In 2026, additional plants were tagged and each blade was punched ~15cm above the pneumatocysts (at T1) to track turn-over (gain and loss) of blades over time. At T2, the number of punched blades was tallied as well as new blades without punches. The same thing was done at T3.

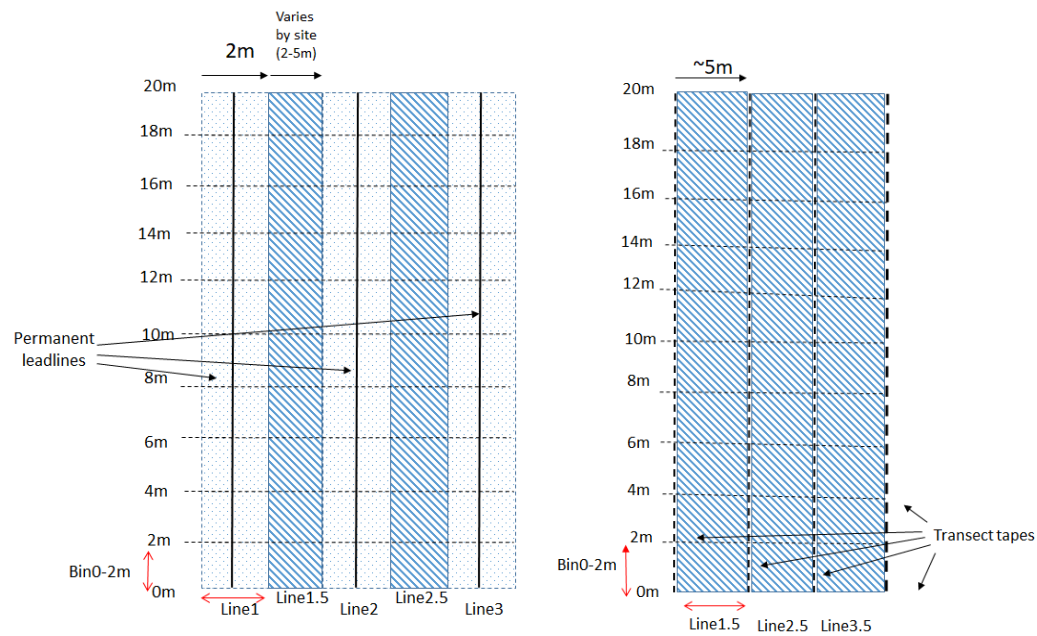


Figure 2. (Left) Example of permanent site layout where additional swaths were surveyed for stipe density between the pre-existing transects (2018 Jul and Aug). (Right) Example of non-permanent site layout where larger swaths were used rather than standard 1m on either side of transect method (2018 Aug Gosling and McMullins).